



Revision



Database Life Cycle

Analysis -> System Analyst

Requirement Document

Database Design -> Database Designer

Entity Relationship Diagram [ERD]

Database Mapping -> Database Designer

Database Schema

Database Implementation -> Database Developer

Using DBMS (SQL Server) To Create The Database

Application -> Application Programmer

Web - Desktop - Mobile

Client -> End User

Use Database Indirectly Through The Application

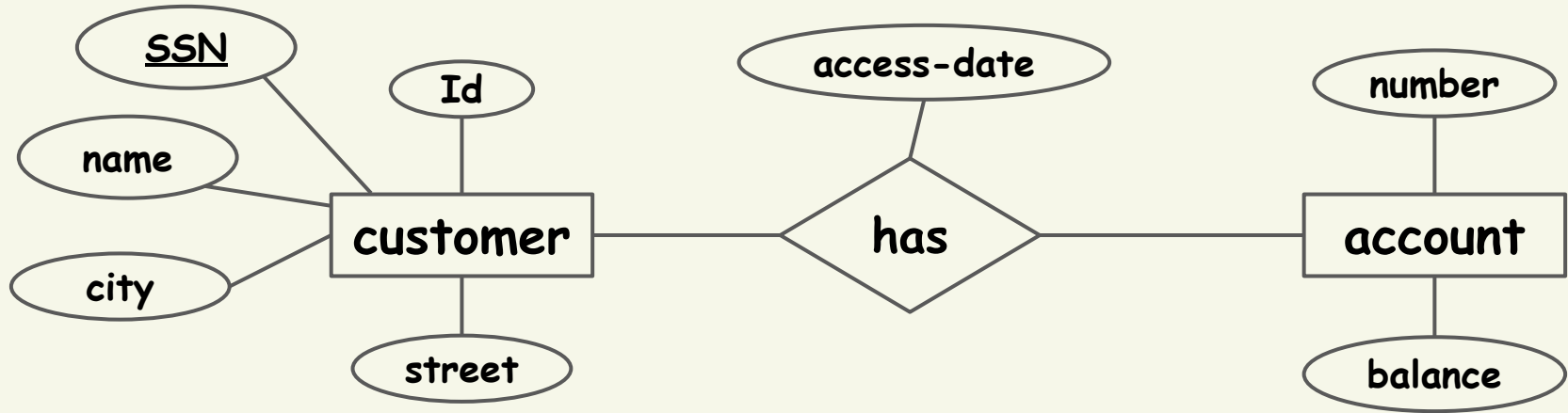
Entity Relationship Diagram Concepts

The ER Model

Basic Components of the E-R model:

- **Entities:** person, place, object, event, concept (often corresponds to a real time object that is **distinguishable** from any other object)
- **Attributes:** property or **characteristic** of an entity type (often corresponds to a field in a table)
- **Relationships:** **link** between entities (corresponds to **primary key - foreign key** equivalencies in related tables)

ER Diagram: Starting Example



- Rectangles: Entity sets
- Ellipses: Attributes
- Diamonds: Relationship sets

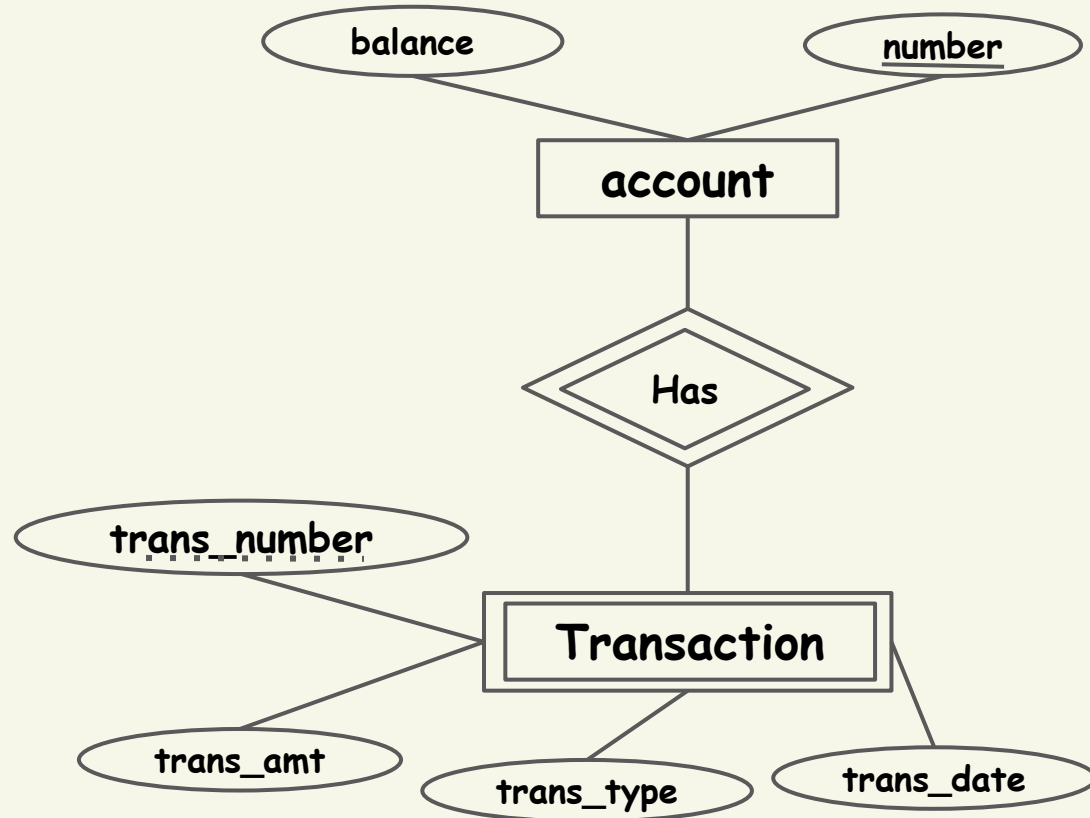
Strong Entity Vs Weak Entity

Strong Entity :

An Entity set that has a primary key.
It doesn't depend on any other entity.

Weak Entity:

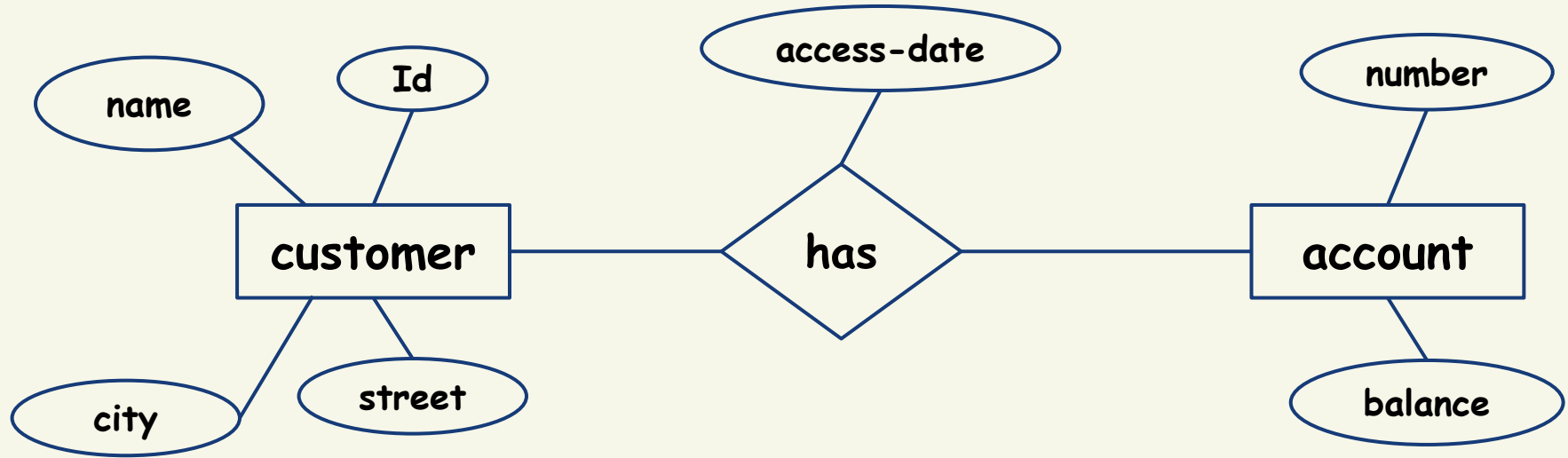
Doesn't have any primary key
because it doesn't have sufficient
attributes to form a primary key
but it has partial key.
It depends on another strong entity



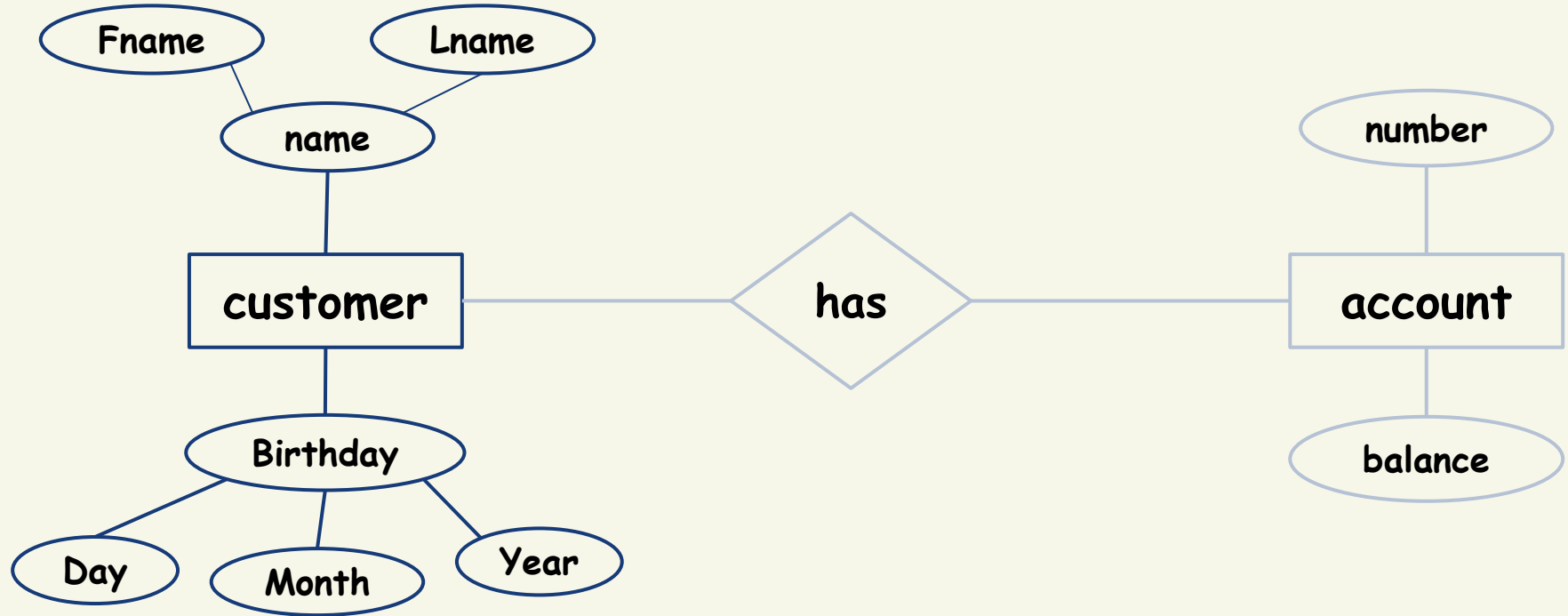
Types of Attributes

1. Simple Attribute
2. Composite Attribute
3. Multi-valued Attribute
4. Derived Attribute
5. Complex Attribute

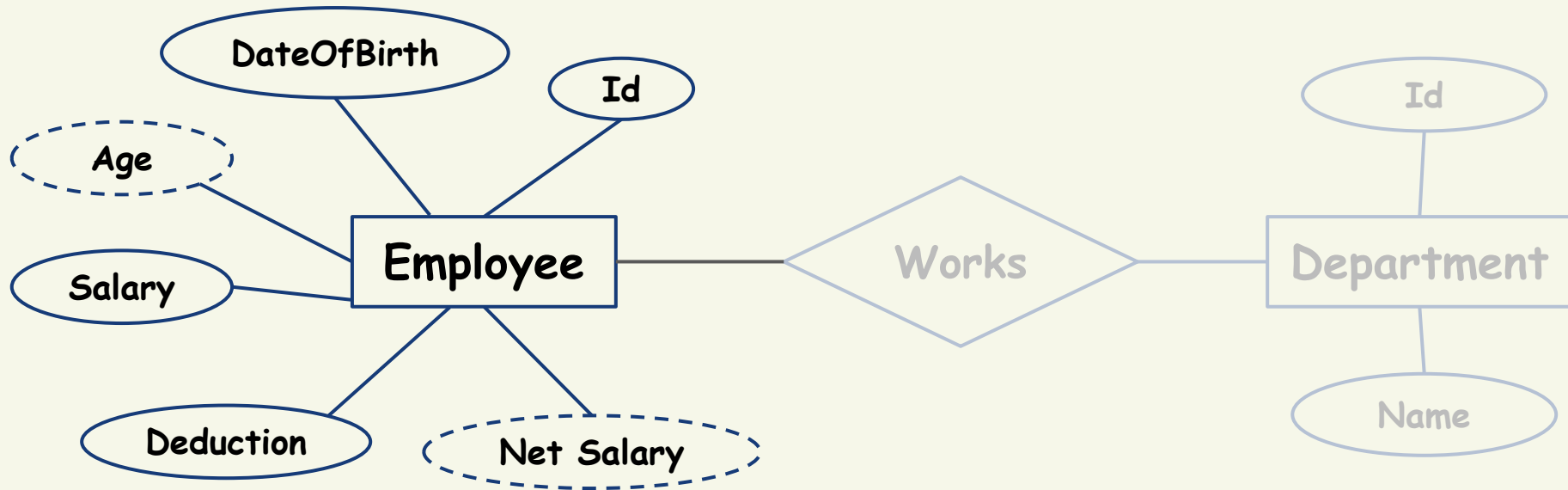
Simple Attribute



Composite Attribute

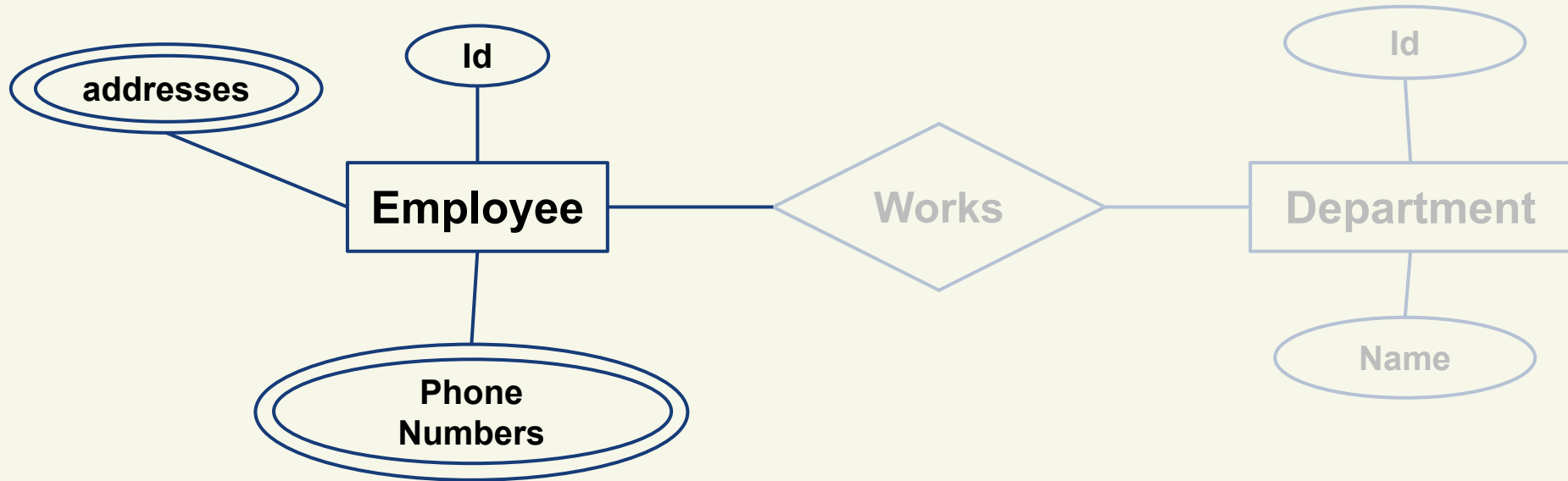


Derived Attribute (dashed ellipse)

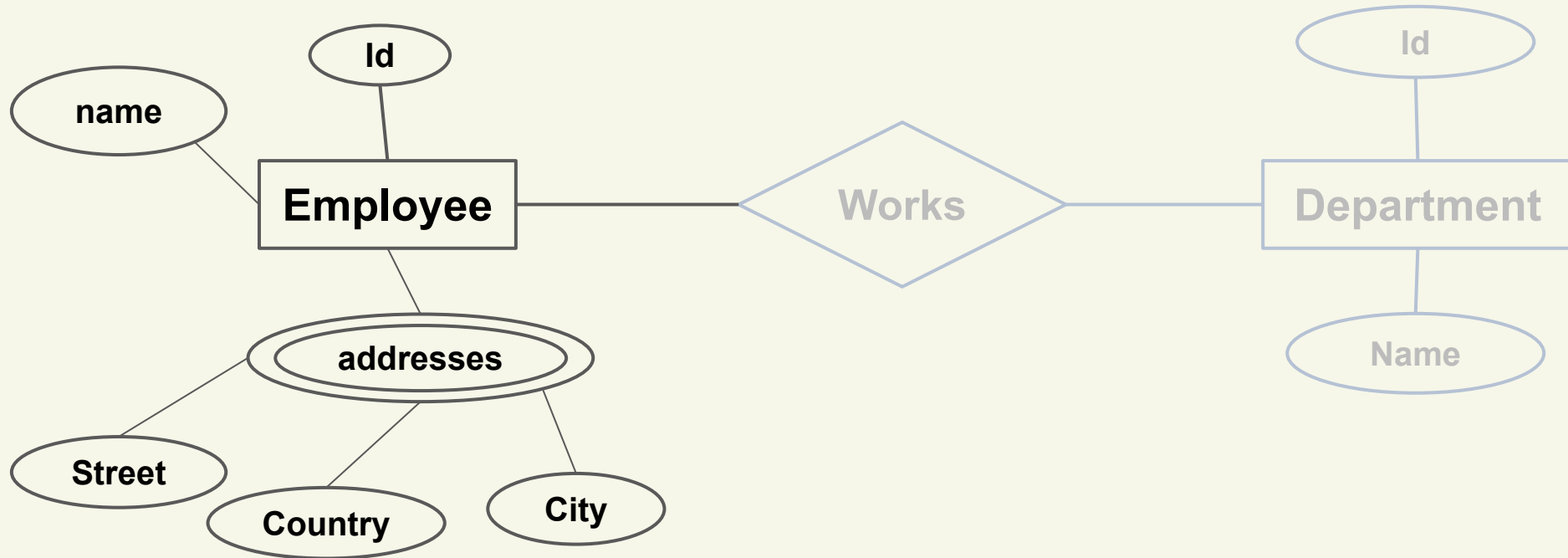


❖ They don't exist physically in the database

Multi-valued Attribute(double ellipse)



Complex Attribute (multi-valued + Composite)



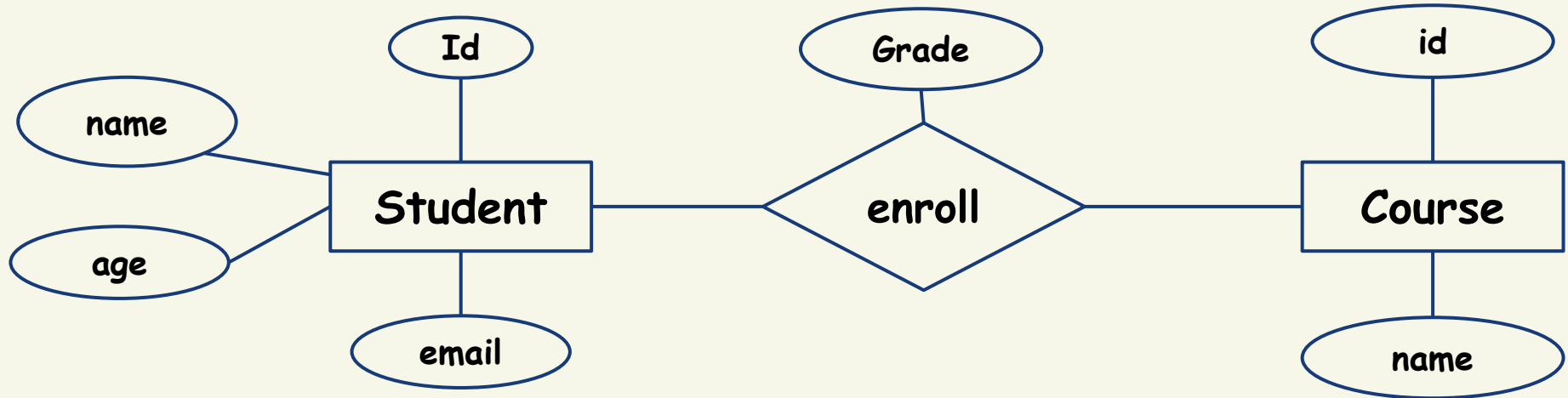
Types of Attributes

1. Simple Attribute
2. Composite Attribute
3. Multi-valued Attribute
4. Derived Attribute
5. Complex Attribute

Relationship

- A Relationship is an association among several entities.
- A relationship may also have attributes

Ex: consider the entity sets student and course and the relationship set enroll. We could associate the attribute **Grade** to that relationship to specify the grade of student in course.



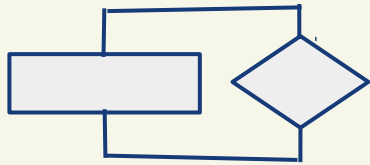
Relationship

Relationship has three Properties:

- Degree of Relationship
- Cardinality Constraint
- Participation Constraint

Degree of Relationships

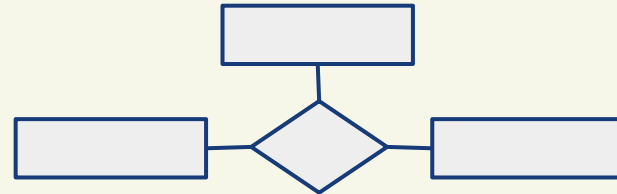
- **Degree:** number of entity types that participate in a relationship
- **Three cases:**
 - **Unary:** between two instances of one entity type
 - **Binary:** between the instances of two entity types
 - **Ternary:** among the instances of three entity types



Unary



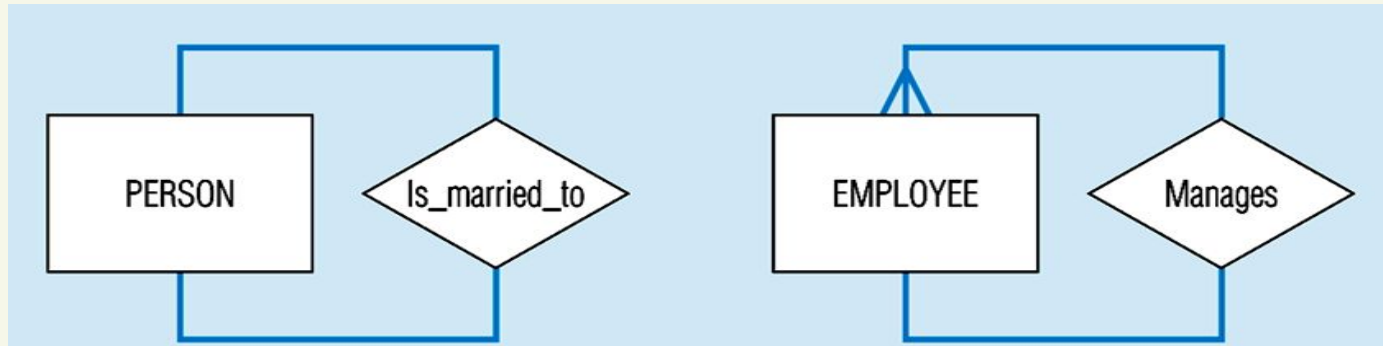
Binary



Ternary

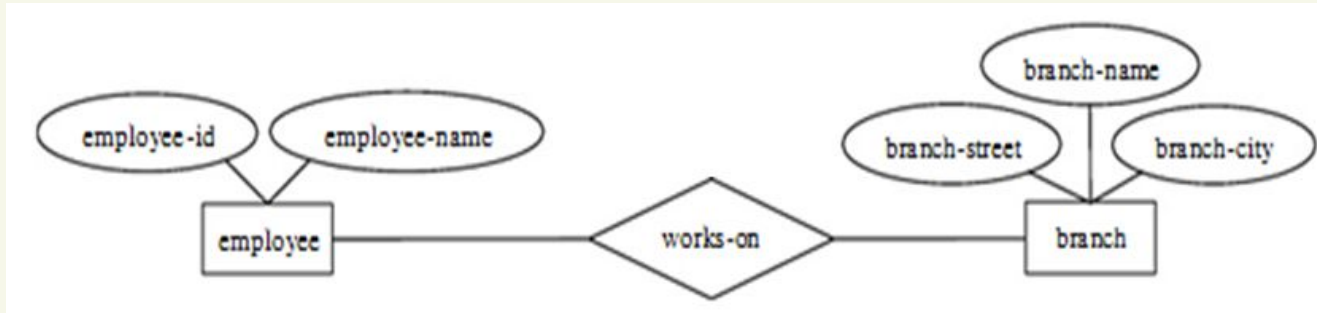
Recursive Relationship (Unary)

Recursive Relationships: A relationship in which the same entity participates more than once.



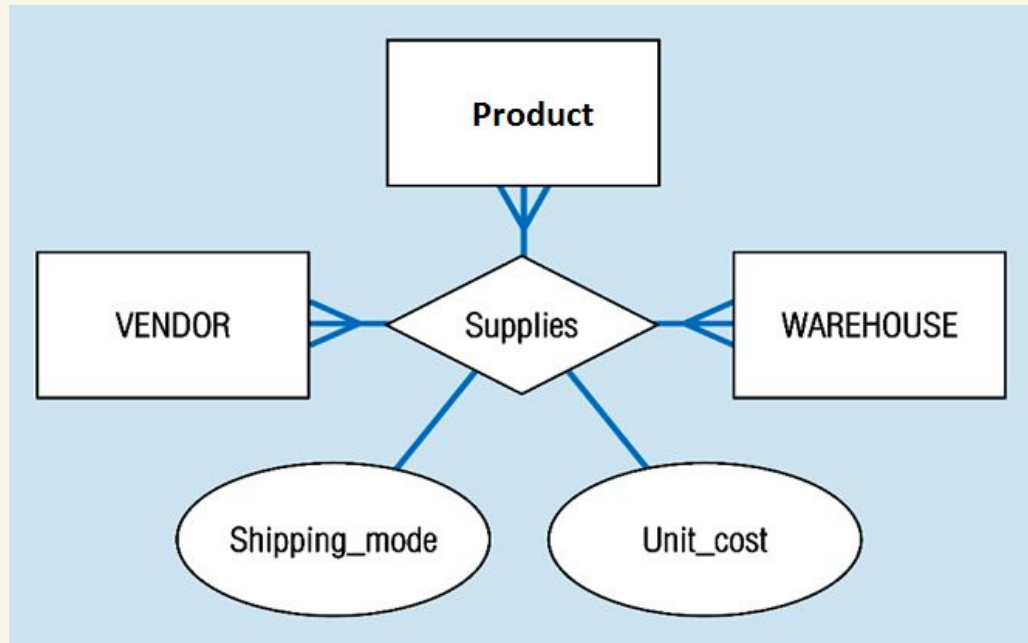
Binary Relationship

A binary relationship set is of degree 2



Ternary Relationship

A Ternary relationship set is of degree 3



Relationship

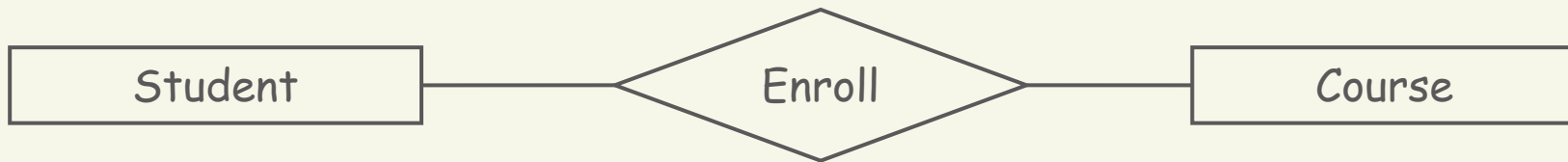
Relationship has three Properties:

- Degree of Relationship
- Cardinality Constraint
- Participation Constraint

Cardinality

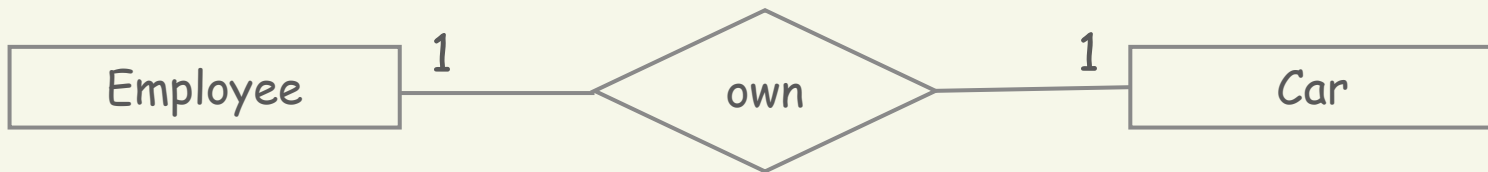
★ How many instances of one entity will or must be connected to a single instance from the other entities:

- **One-One Relationship**
- **One-Many Relationship**
- **Many- Many Relationship**

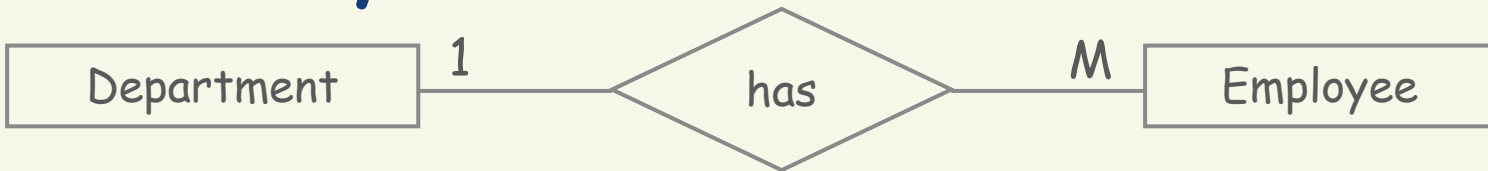


Mapping Cardinalities

□ One-to-One:



□ One-to-Many:



□ Many-to-Many:



Relationship

Relationship has three Properties:

- Degree of Relationship
- Cardinality Constraint
- Participation Constraint

Participation Constraint



An employee **MUST** work for a department :

An employee entity can exist only if it participates in a **WORKS_FOR** relationship instance

So this participation is **TOTAL** (mandatory)



Only some employees manage departments:

The participation is **PARTIAL** (optional)

Participation Constraint



- ❖ An Employee **may** have a car.
- ❖ A Car **must** be assigned to particular employee

Participation Constraint

- ❖ A Department **may** hire many employees (Zero or more)
- ❖ An employee **must** be employed by a department

(Department membership is Optional, Employee membership is Mandatory)





Route
Academy



Database

Session 02



Agenda

- ❑ Relational Database Definitions
- ❑ Mapping Steps
- ❑ Task



Relational Database Definitions

Database : An organized collection of structured information or data, consisting of various database objects such as **tables, views, stored procedures, and triggers**

Relational Database Definitions

Table/Entity : A collection of related data entries
consisting of rows and columns

Attribute/Column/Field : A specific attribute or type of
data stored in a table

Relational Database Definitions

Row/Record/Tuple : A single data item in a table. Each row represents a unique instance of an entity

Primary Key : A column (or set of columns) that uniquely identifies each row in a table

Relational Database Definitions

Foreign Key : A column used to link two tables together, referencing the primary key of another table

Normalization : The process of organizing data to reduce redundancy and improve integrity

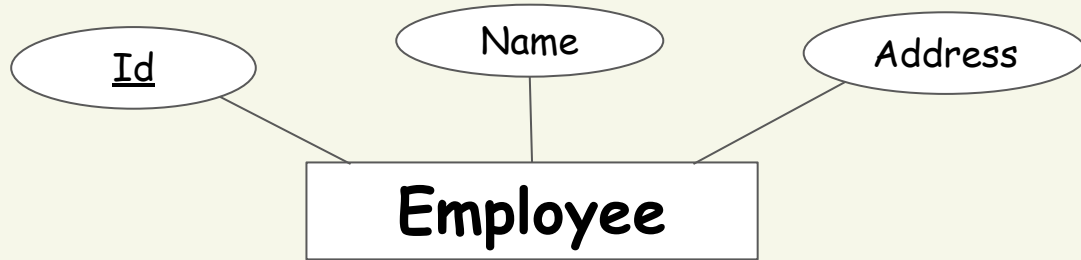
Mapping
SCHEMA

Mapping Steps [Schema]

- Step 0: Mapping 1-1 relationship mandatory from 2 sides.
- Step 1: Mapping of Regular (Strong) Entity Types.
- Step 2: Mapping of Weak Entity Types.
- Step 3: Mapping of Binary 1:1 Relation Types.
- Step 4: Mapping of Binary 1:N Relationship Types.
- Step 5: Mapping of Binary M:N Relationship Types.
- Step 6: Mapping of N-ary Relationship Types.
- Step 7: Mapping of Unary Relationship.

Step 1: Mapping of Regular (Strong) Entity Types

Mapping Simple Attribute:



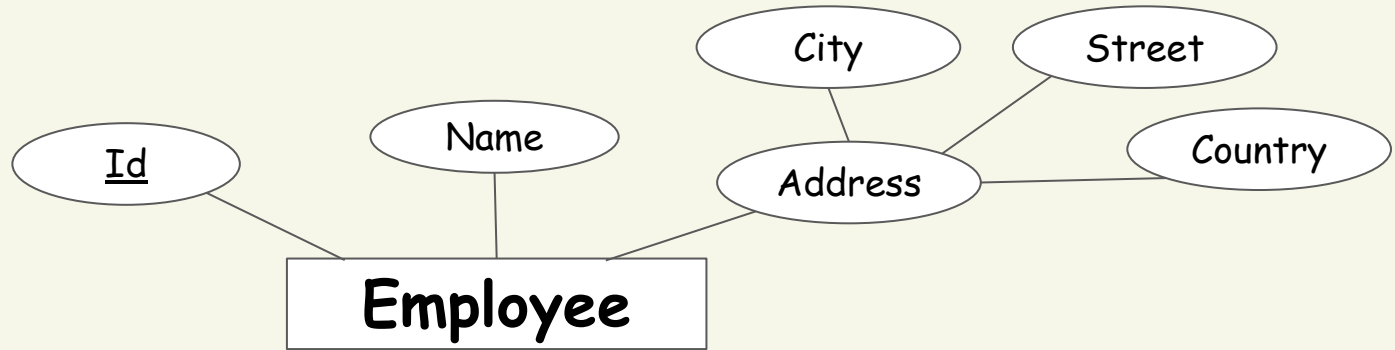
Employee entity type with simple attributes

Employees

<u>Id</u>	Name	Address
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Step 1: Mapping of Regular (Strong) Entity Types

Mapping Composite Attribute:



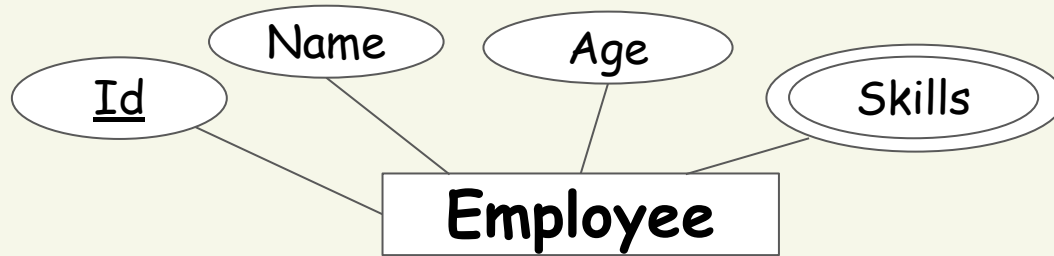
Employee entity type with composite attribute

Employees

<u>Id</u>	Name	City	Street	Country
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Step 1: Mapping of Regular (Strong) Entity Types

Mapping Multivalued Attribute:



Employees

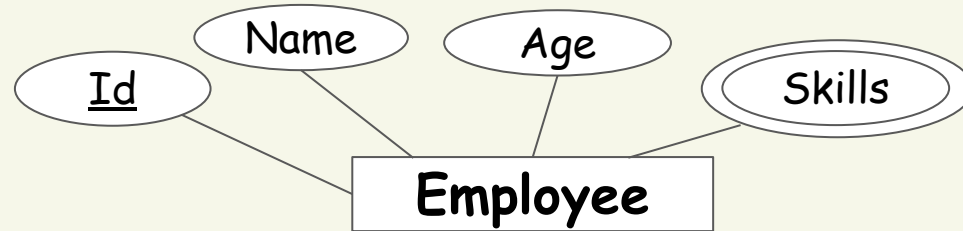
Employee entity type with Multi-Valued attribute

<u>Id</u>	Name	Age
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Employee_Skills

<u>EmpId</u>	<u>Skill</u>
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Step 1: Mapping of Regular (Strong) Entity Types



Employees

<u>Id</u>	Name	Age	Skill
1	Mai	27	C#
2	Amr	25	Java
2	Amr	25	C#
2	Amr	25	C++

Primary Key

Employees

<u>Id</u>	Name	Age
1	Mai	27
2	Amr	25

Foreign Key

Composite PK

Employee_Skill

<u>Emp_Id</u>	Skill
1	C#
2	Java
2	C#
2	C++
10	C#

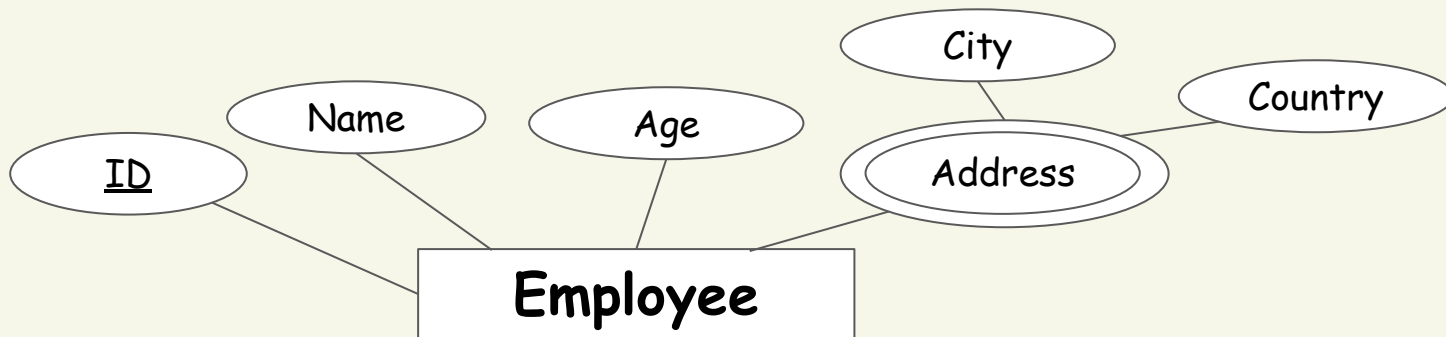
Step 1: Mapping of Regular (Strong) Entity Types

Mapping Derived Attribute:

In the most cases Derived attribute not be stored in DB

Step 1: Mapping of Regular (Strong) Entity Types

Mapping Complex Attribute (Multi-valued + Composite)



Employees

Employee entity type with Complex attribute

<u>Id</u>	Name	Age
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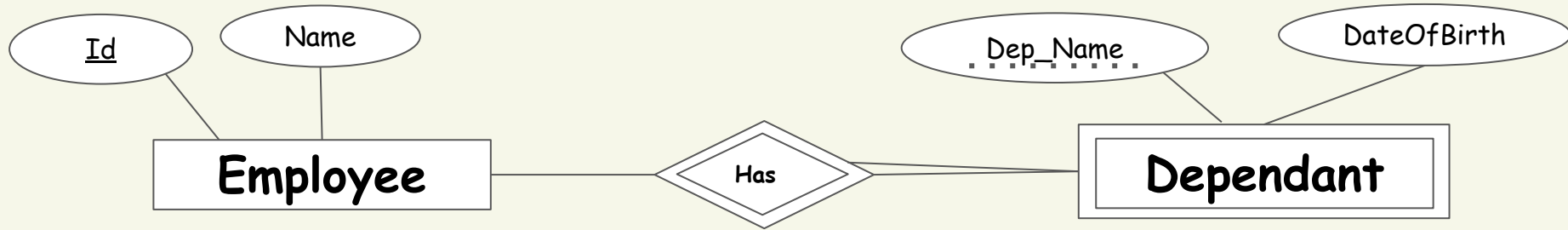
Employee_Addresses

<u>EmpId</u>	City	Country
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Mapping Steps [Schema]

- Step 0: Mapping 1-1 relationship mandatory from 2 sides.
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- Step 3: Mapping of Binary 1:1 Relation Types.
- Step 4: Mapping of Binary 1:N Relationship Types.
- Step 5: Mapping of Binary M:N Relationship Types.
- Step 6: Mapping of N-ary Relationship Types.
- Step 7: Mapping of Unary Relationship.

Step 2: Mapping of Weak Entity Types



Employee entity type in Relationship With Weak entity (Dependant)

Employees

<u>Id</u>	Name
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Dependants

<u>Emp_Id</u>	<u>Dep_Name</u>	DateofBirth
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Composite primary key

Step 2: Mapping of Weak Entity Types

- ❖ Create table for each weak entity.
- ❖ Add foreign key that correspond to the owner entity type.
- ❖ Primary key composed of:
 - Partial identifier of weak entity
 - Primary key of identifying relation (strong entity)

Mapping Steps [Schema]

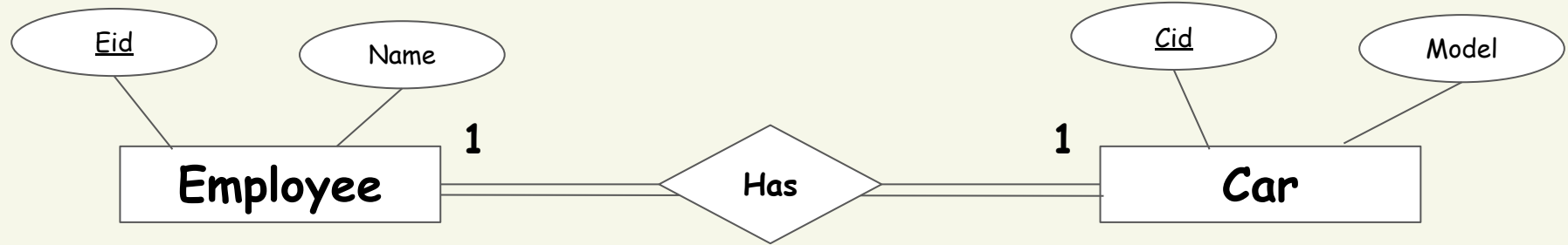
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Step 3: Mapping of Binary 1:1 Relation Types

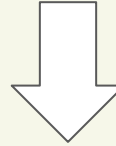
We Have 3 Cases :

1. Both sides are Mandatory [Step 0]
2. Mandatory - Optional
3. Both sides are Optional

Step 0: Mapping 1-1 relationship mandatory from 2 sides



A Mandatory 1-1 Relationship Between Employee and Car

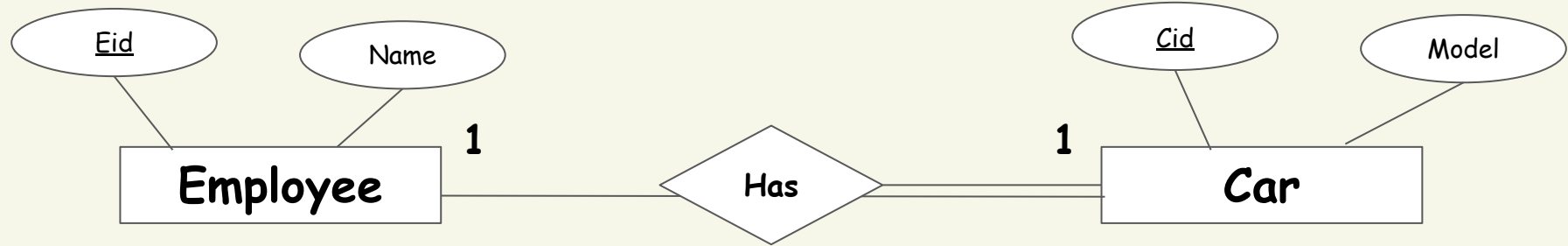


Employees

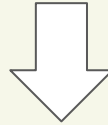
<u>Eid</u>	Name	Cid	Model
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Tablexy : All Attributes of both Entities
PKxy = PKx Or PKy

Step 3: Mapping 1-1 relationship Mandatory - Optional



A Mandatory - Optional 1-1 Relationship Between Employee and Car



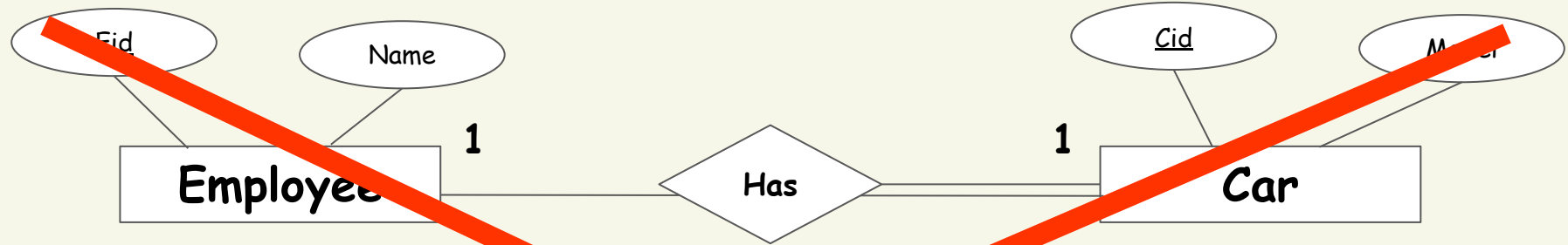
Employees

<u>Eid</u>	Name
------------	------

Cars

<u>Cid</u>	Model	Emp_id
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Step 3: Mapping 1-1 relationship Mandatory - Optional



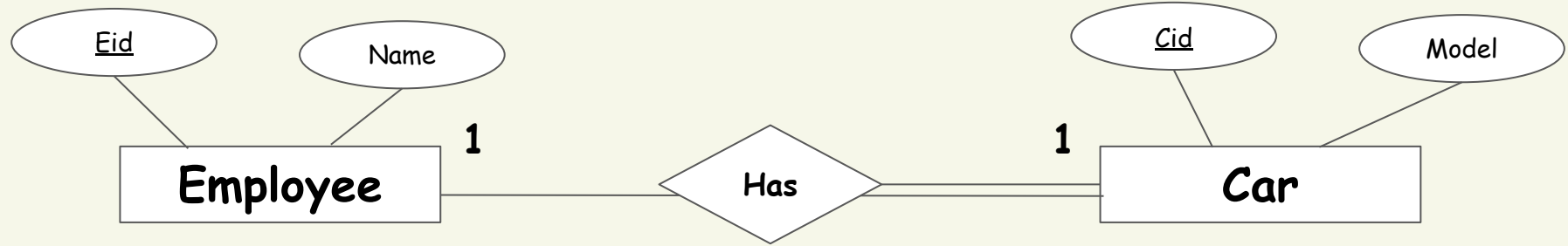
Employees

<u>Eid</u>	Name	Car_Id
1	Amr	10
2	Ali	30
3	Immed	Null
4	Mona	20
5	Salma	Null

Car

<u>Cid</u>	Model
10	BYD
20	AUDI
30	BMW

Step 3: Mapping 1-1 relationship Mandatory - Optional



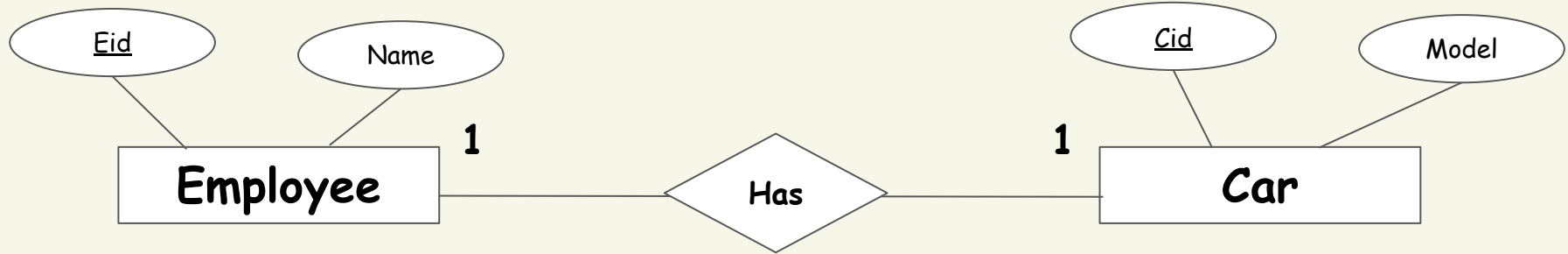
Employees

<u>Eid</u>	Name
1	Amr
2	Ali
3	Ahmed
4	Mona
5	Salma

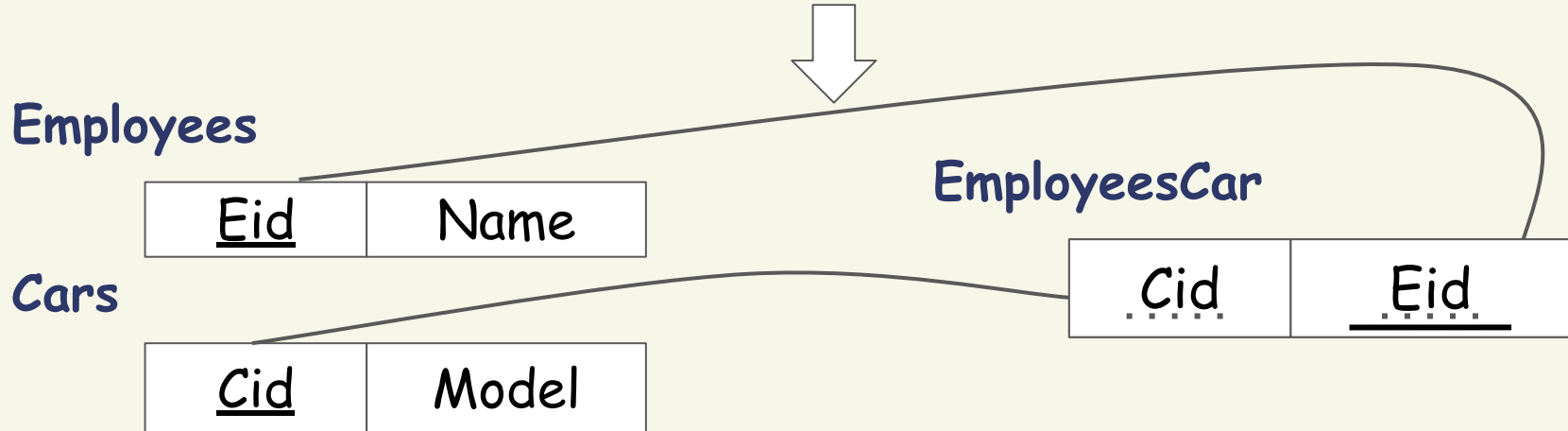
Car

<u>Cid</u>	Model	<u>Emp_Id</u>
10	BYD	1
20	AUDI	4
30	BMW	2

Step 3: Mapping 1-1 relationship Optional - Optional



An Optional 1-1 Relationship Between Employee and Car



Mapping Steps [Schema]

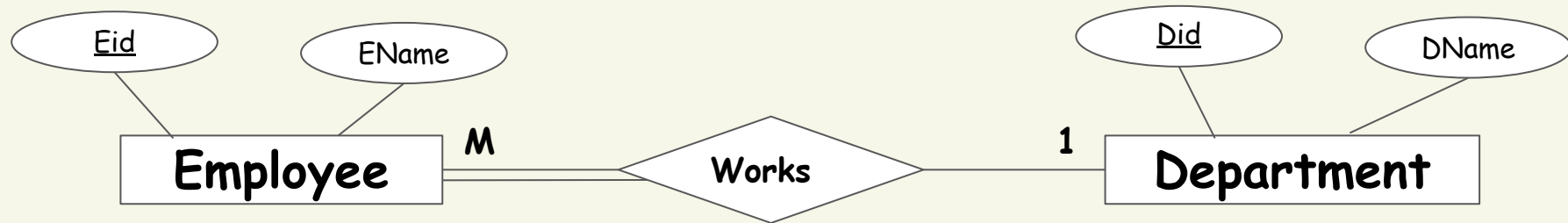
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Step 4: Mapping of Binary 1:N Relationship Types

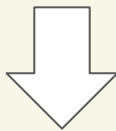
We Have 2 Cases :

1. M-Side is Mandatory
2. M-Side is Optional

Step 4: Mapping of Binary 1:M [M-side Mandatory]



1-M Relationship Between Employee and Department



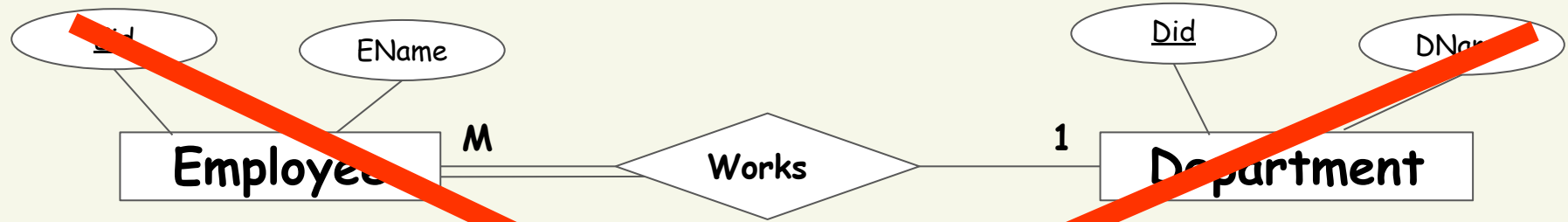
Employees

<u>Eid</u>	EName	Dept_Id
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Departments

<u>Did</u>	DName
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Step 4: Mapping of Binary 1:M [M-side Mandatory]



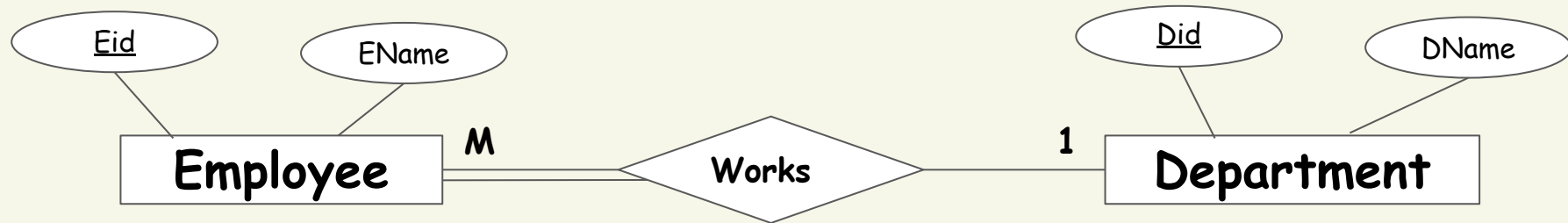
Employees

<u>Eid</u>	Name
1	Amr
2	Ali
3	Alihed
4	Mona
5	Salma

Department

<u>Did</u>	DName	Emp_Id
100	HR	1
200	IT	4
300	Sales	2
200	IT	3
200	IT	5

Step 4: Mapping of Binary 1:M [M-side Mandatory]



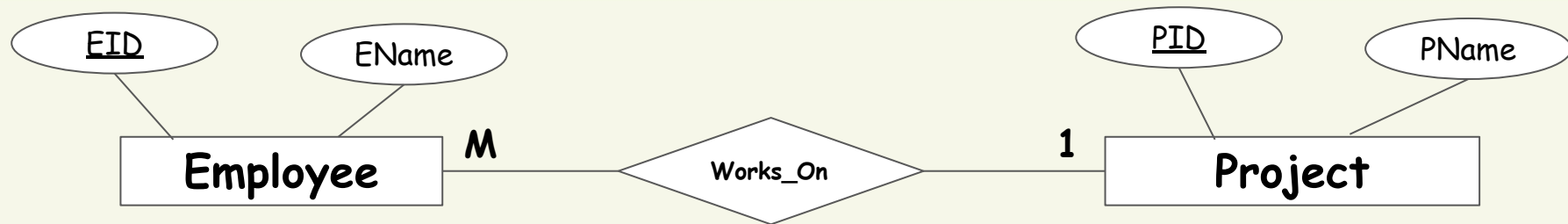
Employees

<u>Eid</u>	Name	Dept_Id
1	Amr	100
2	Ali	300
3	Ahmed	200
4	Mona	200
5	Salma	200

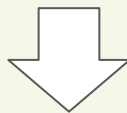
Department

<u>Did</u>	DName
100	HR
200	IT
300	Sales

Step 4: Mapping of Binary 1:M [M-side Optional]



1-M Relationship Between Employee and Project



Employees

<u>Eid</u>	EName
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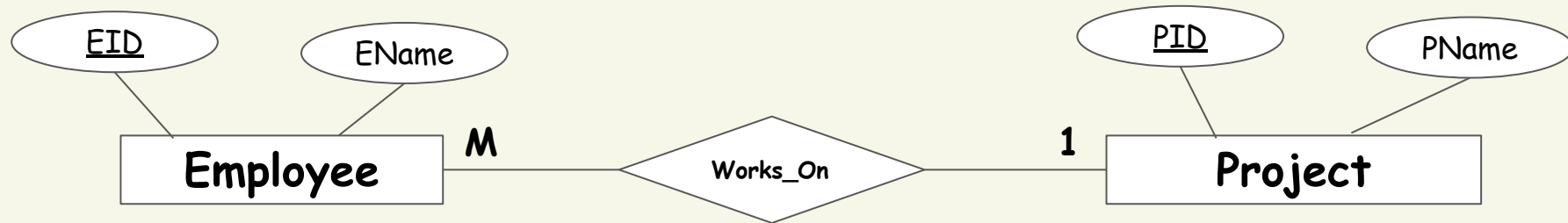
Project_Employees

.. Pid ..	<u>.. Eid ..</u>
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Projects

<u>Pid</u>	PName
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Step 4: Mapping of Binary 1:M [M-side Optional]



Employees

<u>Eid</u>	EName
1	Amr
2	Ali
3	Ahmed
4	Mona
5	Salma

Projects

<u>Pid</u>	PName
100	X
200	y

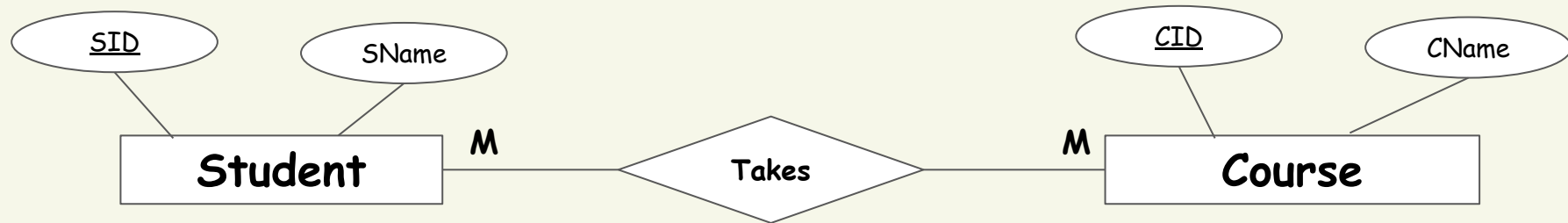
Project_Employees

<u>Pid</u>	<u>Eid</u>
100	1
200	2
200	5
200	4
100	3

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Step 5: Mapping of Binary M:N Relationship Types



M-M Relationship Between Student And Course



Students

<u>Sid</u>	SName
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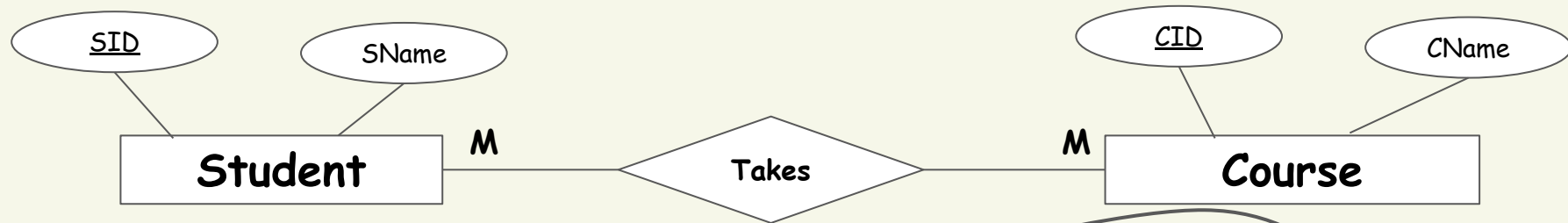
Courses

<u>Cid</u>	CName
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Student_Courses

<u>. . . Cid . . .</u>	<u>. . . Sid . . .</u>
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Step 5: Mapping of Binary M:N Relationship Types



Students

<u>Sid</u>	EName
1	Manar
2	Samy
3	Shaimaa
4	Youstina
5	Ahmed

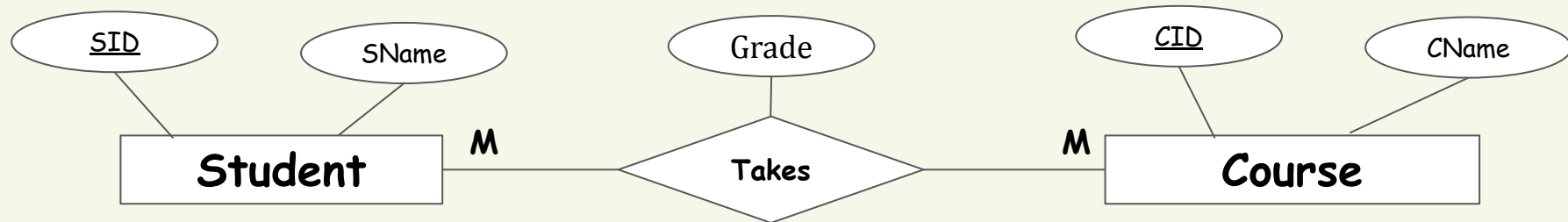
Courses

<u>Cid</u>	CName
10	Frontend
20	Backend

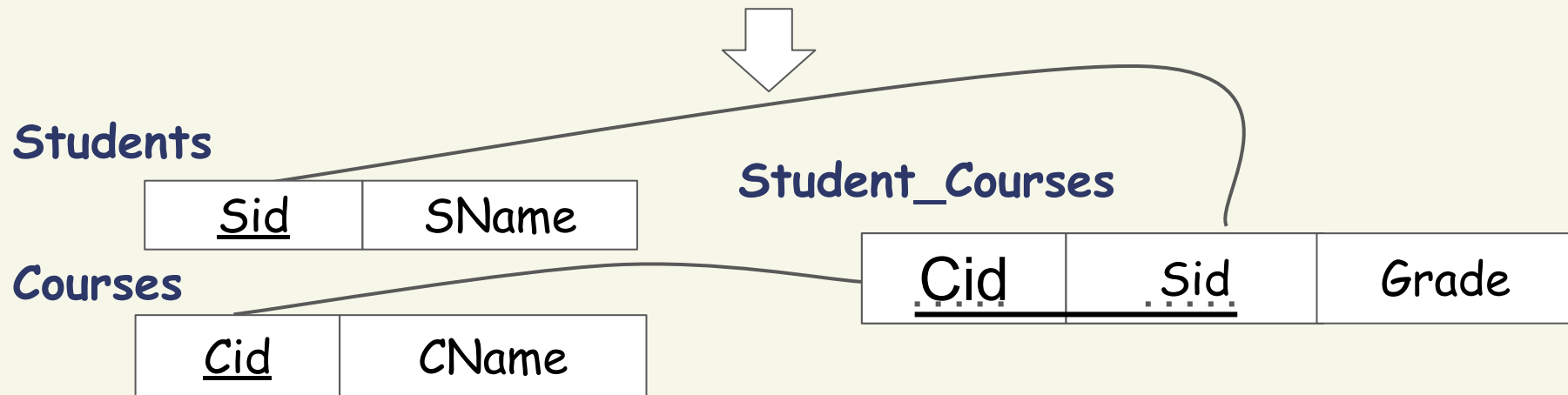
Student_Courses

<u>Sid</u>	<u>Cid</u>
1	10
2	10
3	20
1	20
4	10

Step 5: Mapping of Binary M:N Relationship Types



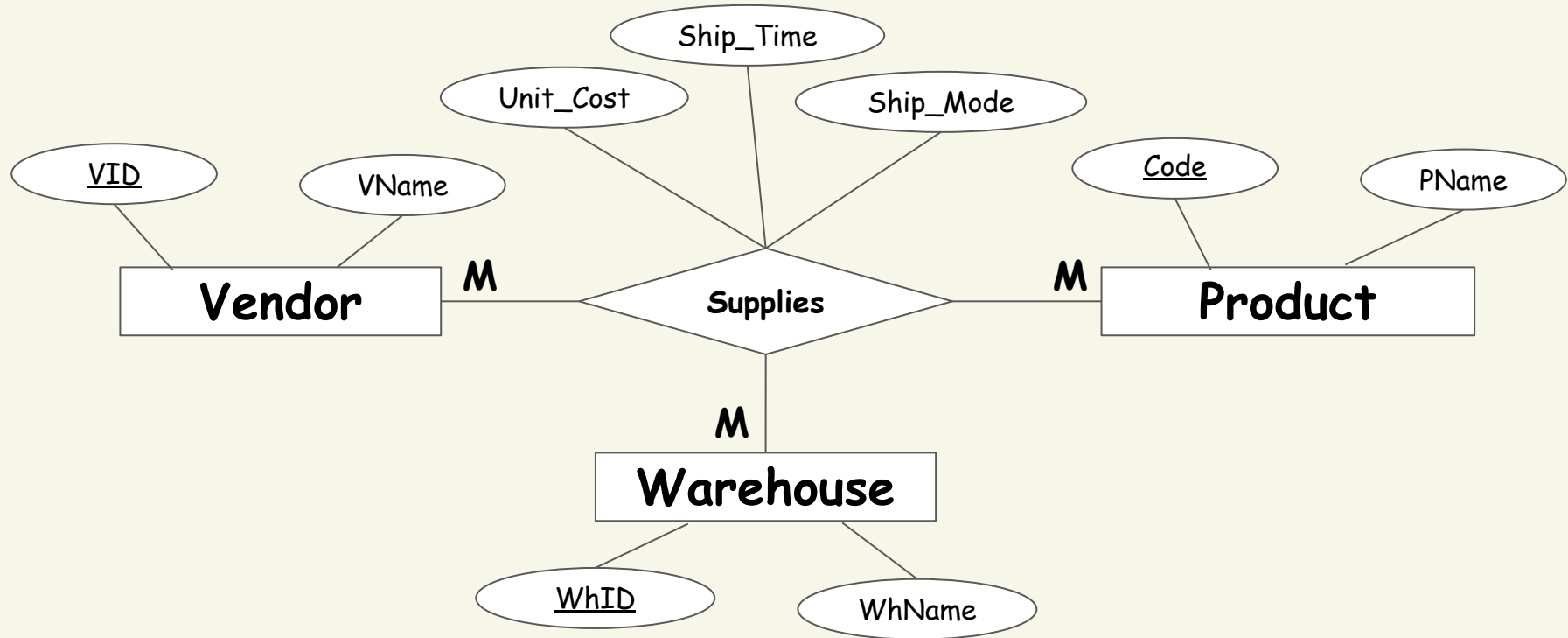
M-M Relationship With Attribute On Relation Between Student And Course



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- Step 7: Mapping of Unary Relationship.

Step 6: Mapping of N-ary Relationship Types



Ternary Relationship With Attribute On Relation Between Vendor, Product and Warehouse

Step 6: Mapping of N-ary Relationship Types

Vendor

<u>Vid</u>	VName
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Warehouse

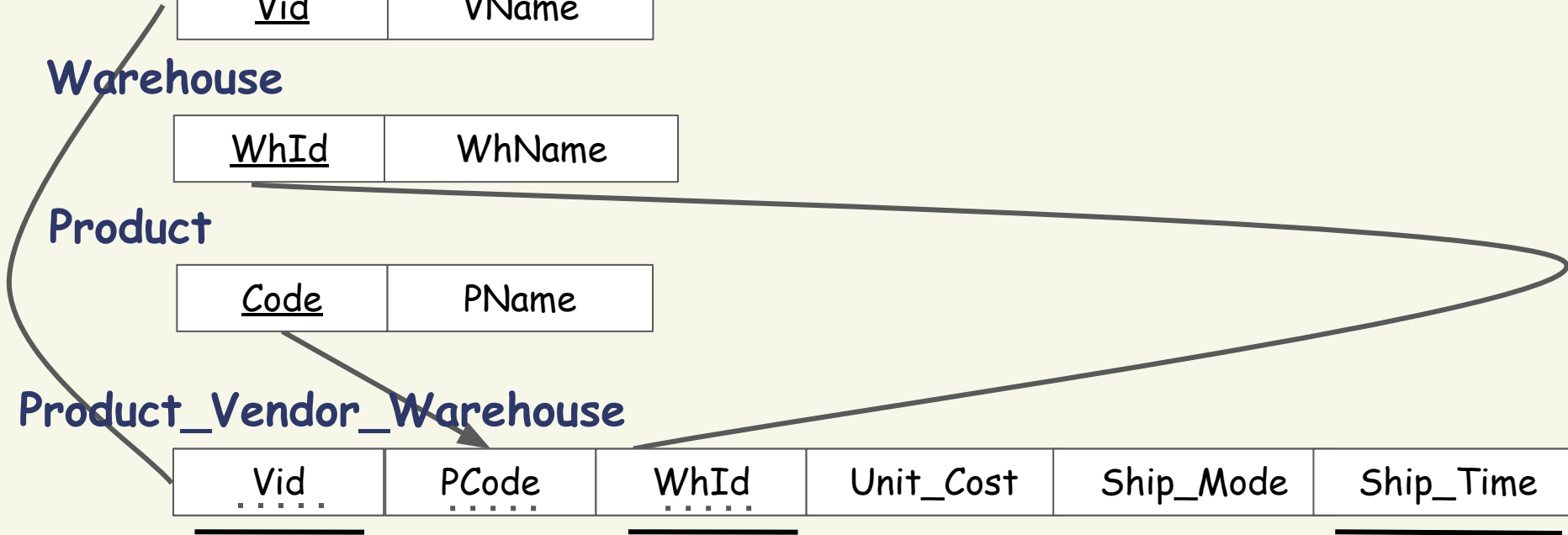
<u>WhId</u>	WhName
-------------	--------

Product

<u>Code</u>	PName
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Product_Vendor_Warehouse

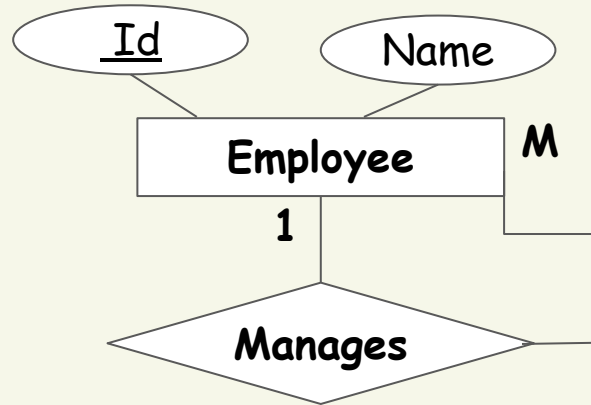
<u>Vid</u>	PCode	<u>WhId</u>	Unit_Cost	Ship_Mode	Ship_Time
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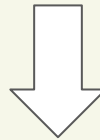
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- Step 6: Mapping of N-ary Relationship Types.
- Step 7: Mapping of Unary Relationship.

Step 7: Mapping of Unary Relationship



Unary Relationship Between 2 Instances From Employee Entite



Employees

<u><i>Id</i></u>	<i>Name</i>	<i>Manager_Id</i>
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Mapping Steps [Schema]

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- Step 3: Mapping of Binary 1:1 Relation Types.
- Step 4: Mapping of Binary 1:N Relationship Types.
- Step 5: Mapping of Binary M:N Relationship Types.
- Step 6: Mapping of N-ary Relationship Types.
- Step 7: Mapping of Unary Relationship.

Relational Mapping (Schema) Summary

Attributes Mapping

Simple : Directly map as a column in the table

Composite : Break into individual components; map each component as a separate column

Derived : Do not store in the table; compute as needed using queries or views

Multivalued : Create a separate table. Include:

- A foreign key referencing the parent entity
- The multivalued attribute as a column
- Composite primary key (foreign key + attribute)

Relational Mapping (Schema) Summary

1 - 1 : One Table , Choose PK of any entity to this table

1 - 1 : 2 Tables , Take PK Of Optional As FK In Mandatory Table.

1 - 1 : 3 Tables , Choose PK of any entity to this table.

1 - M : 2 Tables , Take PK Of One As FK in Many

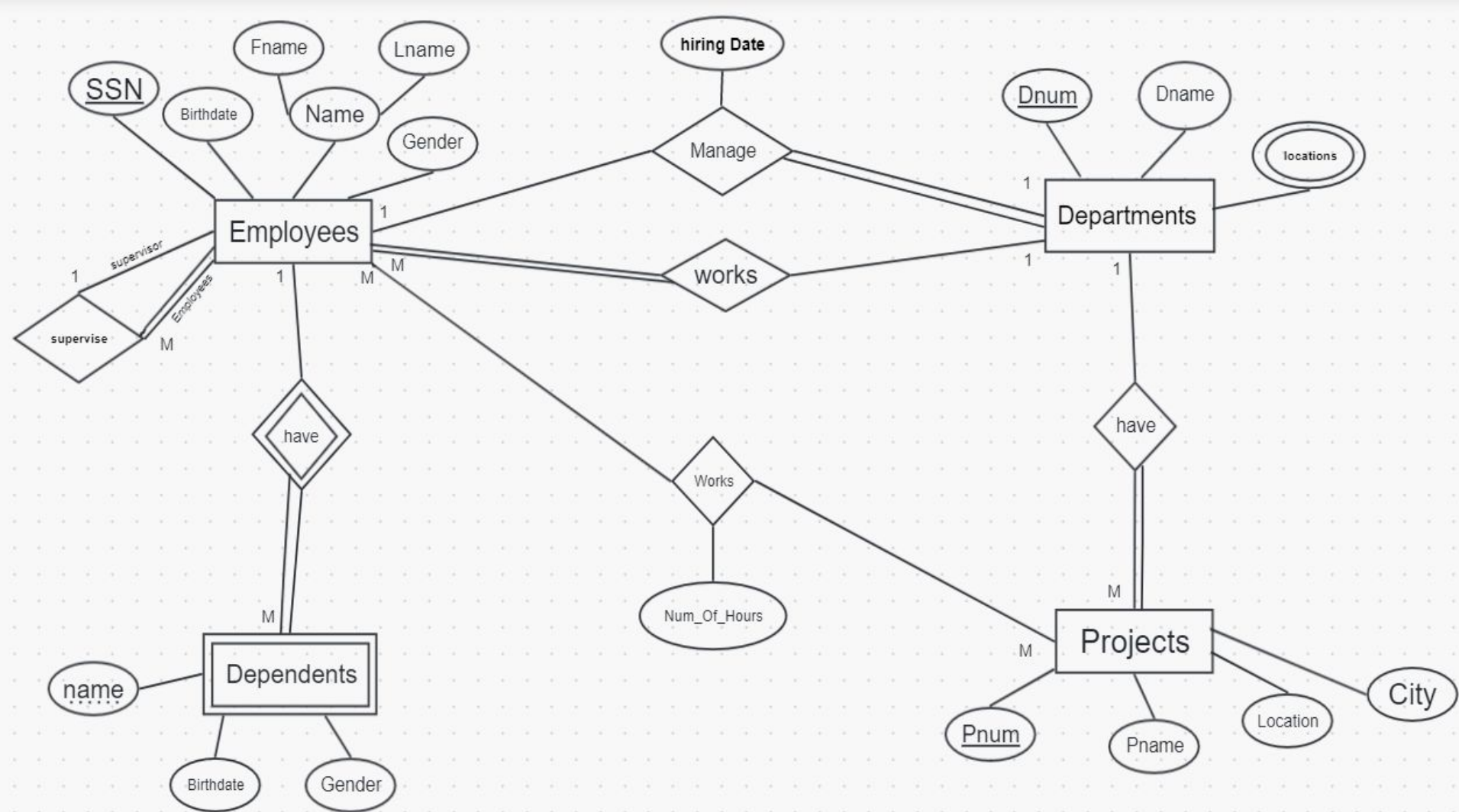
1 - M : 3 Tables , PK of Third Table is PK of Many

M - M : 3 Tables , PK of Third Table is Composite PK (PK1 + PK2)

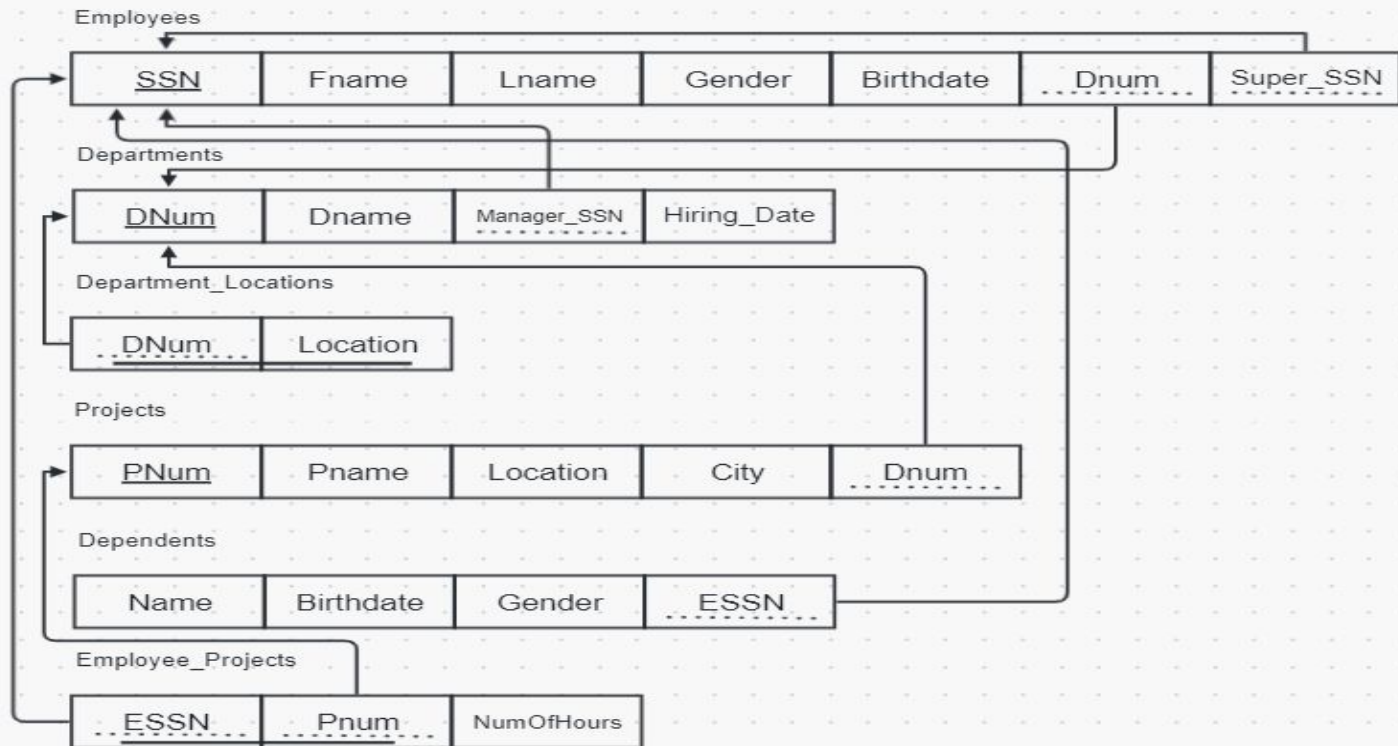
Ternary : Table for Each Entity + Table for Relation Contains PKs of each table as FK (PK is Composite PK).



CASE STUDY



MAPPING RESULT





Route
Academy

Thank You

See You Next Session

